Finite-Difference Certificates and a Phase-Separation Characterisation for Erdős Problem 249

A companion note on transport, curvature, and local obstructions

Will Cook

July 2026

Abstract

Let $S = \sum_{n\geq 1} \varphi(n)/2^n$ and $R_N = \sum_{j\geq 1} \varphi(N+j)/2^j$. The Möbius residue kernel governing transport of the tails R_N is affine in its scale coordinate, with periodic slope and intercept. Consequently, every finite coefficient system satisfying $\sum_i c_i = \sum_i c_i m_i = 0$ annihilates all divisor channels already present at a base height. Curvature, prime-jump, balanced $(1,2,3,6)$, and anchored $(1,3,5,15)$ observables admit exact dyadic-window decompositions and decidable central-residue conditions implying non-integrality. A first-harmonic saving on a finite block forces a tail-difference certificate, and the fibre-free cofinal window-pair separation predicate is equivalent to the irrationality of S. We also prove two local obstructions: every fixed-precision valuation–unit word has a prefix-locked centred completion, and a synthetic nonzero lcm carry satisfying the stated factor and anchor constraints survives every finite integer shift polynomial. The finite criteria and obstruction theorems are unconditional. Erdős Problem 249 remains open because no cofinal arithmetic or harmonic producer is proved.

1 Main results and logical boundary

The gateway exposition [7] proves the denominator exclusion and the complete finite-certificate normal form for S. This paper has a different thesis: finite differences annihilate the old affine channels exactly, phase separation gives a complete geometric normal form, and two countermodels delimit what fixed local transport data can prove.

Theorem 1.1 (affine-channel annihilation). *Let $m_i \in \mathbb{N}$ and $c_i \in \mathbb{R}$ satisfy $\sum_i c_i = 0$ and $\sum_i c_i m_i = 0$. If $d \mid H$, then*

$$\sum_i c_i K_d(m_i H) = 0,$$

where K_d is the exact Möbius transport-residue kernel.

Formalised as affine-channel annihilation. The anchored coefficients at $H, 3H, 5H, 15H$ are one primitive instance.

Theorem 1.2 (phase-separation characterisation). *The cofinal fibre-free window-pair separation predicate of Section 8 holds if and only if S is irrational.*

Formalised as the phase-separation equivalence. This is an exact equivalence, not a weaker sufficient condition. Its arithmetic producer remains open.

Proposition 1.3 (local no-go boundaries). *At every fixed valuation–unit precision, each compatible finite word has a prefix-locked centred carry completion. Separately, for every lcm height $t \geq 3$, a nonzero synthetic carry satisfies the stated factor ideals, whole-ray anchors, and every finite integer shift polynomial while remaining a strict survivor.*

Formalised by the fixed-precision countermodel and the finite shift-algebra countermodel. These are scoped countermodels, not global impossibility theorems.

Every positive endpoint below still has the form

$$\text{cofinal supply of finite certificates} \Longrightarrow S \notin \mathbb{Q}.$$

The finite identities, criteria, equivalence, and countermodels are unconditional; no cofinal supply is proved. The open problem is recorded in the original collection and maintained catalogue [4, 5, 6]. The Lambert-series and Möbius/divisor background is classical [1, 2]; Borwein's theorem gives a neighbouring general irrationality benchmark [3]. None supplies the cofinal producer isolated here. The theorem-specific comparison boundary is maintained with the gateway paper. Linked mathematical phrases open the pinned Lean proofs; the claim registry and release gate that validate these links are described in the suite's systems companion [8].

2 Dyadic windows

For a finite integer combination $X(H) = \sum_i c_i R_{m_i H}$, truncate each tail at depth L. Multiplication by 2^L gives

$$2^L X(H) = A(H, L) + E(H, L),$$

where $A(H, L) \in \mathbb{Z}$ is computable. If $|E(H, L)| \leq B(H, L)$, then $X(H) \notin \mathbb{Z}$ whenever

$$B(H, L) < A(H, L) \bmod 2^L < 2^L - B(H, L).$$

This reusable test is the formal central-residue criterion. Lean checks the exact decomposition, radius, and concrete modular arithmetic; an external analytic or arithmetic argument must provide unbounded instances.

3 Curvature on the lcm cone

Set

$$Q_H = R_{4H} - 3R_{2H} + 2R_H.$$

Its depth-L window $C(H, L)$ satisfies

$$2^L Q_H = C(H, L) + E_C(H, L), \qquad |E_C(H, L)| \leq 6H + 3L + 3.$$

The exact split and the tail bound are both checked in Lean: exact curvature split and curvature tail bound. The sharp curvature criterion therefore proves $Q_H \notin \mathbb{Z}$. An unbounded supply of these certificates would imply irrationality by the curvature-supply implication, but that supply is not proved.

The window evolves by $C(H, L + 1) = 2C(H, L) + d(H, L + 1)$; see the exact curvature recurrence. Exact phase-certificate and common-carrier formulas expose possible routes to a supply theorem without asserting one.

4 Exponent transport and affine cancellation

If $p \mid H$, multiplication by p adds no squarefree divisor support:

$$d \mid pH \Longleftrightarrow d \mid H \qquad (d \text{ squarefree}).$$

This is the formal squarefree support invariance. The exact residue kernel is affine, not merely periodic:

$$K_d(N) = N\, a_d(N) + b_d(N),$$

where both a_d and b_d are periodic modulo d, as recorded in the affine transport-kernel theorem. Keeping both blocks prevents an intercept-only matrix calculation from being misread as global non-cancellation. The exact $H \mapsto 3H$ two-block transfer is also formalised by the three-scale transfer identity.

Every old channel $d \mid H$ is killed by the first-order defect in the old-channel cancellation theorem. The resulting irrationality theorem remains conditional on a fixed-three tower supply; see the exponent-transport supply implication.

More generally, coefficients satisfying $\sum_i c_i = \sum_i c_i m_i = 0$ annihilate every old affine channel by affine moment annihilation. For $q(X,Y) = XY - 3X - 2Y + 4$, the vanishing $q(1,1) = q(3,5) = 0$ gives

$$K_d(15H) - 3K_d(3H) - 2K_d(5H) + 4K_d(H) = 0 \quad (d \mid H),$$

as checked in Lean by anchored $(3,5)$ cancellation. The associated window radius is $19H+5L+5$; it gives a finite non-integrality criterion in the joint-transport certificate. The corresponding irrationality endpoint is conditional on the anchored joint-cone supply.

5 Prime jumps and three-transport boundaries

When $p \nmid H$, a jump from H to pH creates new squarefree support. The residue Hecke defect splits into a post-jump channel and an explicit migration correction by the prime-jump migration identity. For $d \mid H$ the total old-channel defect is zero by prime-jump old-channel cancellation; the $(d, H, p) = (5, 12, 5)$ calculation records the two cancelling terms.

Set

$$J(H, p) = (R_{2pH} - R_{pH}) - p(R_{2H} - R_H).$$

Its exact remainder radius is $3pH + (p + 1)(L + 2)$, with a checked window split in the prime-jump window identity. This yields a finite non-integrality criterion from a sharp prime-jump certificate. Lean verifies the instance $(H, p, L) = (12, 5, 15)$ in the $(12, 5, 15)$ calculation; this is one finite deposit, not an unbounded sequence.

The balanced vertices $H, 2H, 3H, 6H$ give a related radius $9H + 4L + 4$. The corresponding finite criterion is formalised as the three-transport certificate; the irrationality theorem is conditional on an unbounded supply through the roughness-boundary implication. The concrete instance $(H, L) = (60, 12)$ is checked in the $(60, 12)$ calculation.

6 A first-harmonic criterion

For the window discrepancy $A(h, N, L)$ define

$$W_1(h, N, L) = \cos\left(2\pi \frac{A(h, N, L) \bmod 2^L}{2^L}\right).$$

Assume $N \in [X, 2X)$ and $16(2X + h + L + 2) \leq 2^L$. If every N failed the central-residue certificate, each phase would lie within $\pi/8$ of an integer turn, so each cosine would exceed $9/10$. Hence

$$\sum_{X \leq N < 2X} W_1(h, N, L) \leq \frac{9}{10} X$$

forces a certificate in the block by the block first-harmonic criterion. More generally, if T is any nonempty finite set of indices below $2X$ and

$$\sum_{N \in T} W_1(h, N, L) \leq \frac{9}{10} |T|,$$

then a certificate occurs at some $N \in T$ by the finite-subset criterion. By certificate completeness, the witness produced here is an exact non-integral tail difference, not a heuristic phase test. Hence a first-harmonic saving on one finite block produces exactly the kind of witness required by the tail-difference criterion. What is still missing is a first-harmonic estimate that supplies such blocks at unbounded parameters; no such estimate is proved here.

7 Adjacent-pair phase separation

At a scale satisfying the same room inequality, with $N+1 < 2X$, suppose that the first harmonic phases at two adjacent indices obey

$$\|\mathrm{windowFirstExp}(h, N+1, L) - \mathrm{windowFirstExp}(h, N, L)\|^2 \geq \frac{19}{25}.$$

Then a certificate occurs at one of N and $N+1$ by adjacent-phase separation. Consequently, a cofinal supply of adjacent pairs satisfying this inequality implies irrationality of the totient constant through the adjacent-phase supply implication. No cofinal adjacent-pair separation is proved here.

8 Pivot anti-reconstruction and window-pair geometry

The first-harmonic phase of a finite window is the exact additive character of its dyadic discrepancy residue. It differs from the corresponding infinite tail-orbit phase by the explicitly bounded omitted tail. On a finite supplier fibre, the anchored defect is the squared distance of the window phases from the constant phase 1; its centred form is the usual pairwise-energy identity. Thus a counted family of separated phase pairs of sufficient total squared chord mass produces a finite certificate.

The formal proof records several equivalent sufficient cofinal hypotheses. In particular, a cofinal pivot anti-reconstruction bound, a cofinal centred reconstruction-variance bound, or a cofinal counted-separated-pairs bound is sufficient for irrationality of S. The direct finite criterion and the resulting conditional irrationality theorem are both formalised by the pivot-fibre certificate and the pivot anti-reconstruction implication.

There is also an exact fibre-free formulation: the cofinal window-pair predicate is equivalent to irrationality of S, not merely sufficient for it. The reverse implication uses the expansive doubling orbit to obtain adjacent infinite-tail separation and then chooses a finite window depth in the exact window-pair equivalence. The development does not prove any of these cofinal arithmetic hypotheses, so it does not settle Erdős #249.

9 Fixed local precision cannot finish the argument

A finite valuation–unit symbol records a bounded amount of local transport data. For every positive precision u, every finite compatible word admits a prefix-locked centred completion. Thus no finite word over this fixed local alphabet excludes all centred endpoints by the fixed-precision no-go theorem. This does not say curvature certificates are rare. A proof may use precision growing with scale, global correlations, or a harmonic gap. It rules out only the inference that a bounded local signature alone forces escape.

10 LCM factor ideals and whole-ray anchors

For every $t \geq 3$, there is a nonzero synthetic affine carry whose forcing letters lie in the principal ideal generated by $\varphi(\mathrm{lcm}(1, \ldots, t))$, satisfy the exact whole-ray anchor values, and remain inside the relevant diagonal bounds. Nevertheless its state is a strict survivor in the sparse-anchor countermodel.

The same example remains a dyadic coboundary after every finite integer shift polynomial. In particular, the finite-rank LCM shift algebra alone cannot remove its terminal compensation in the finite shift-algebra countermodel. This is a synthetic countermodel: it does not assert that these forcing letters are actual totient differences, and it does not rule out a nonlinear argument using the missing fresh divisor channels.

11 Statement matrix and open boundary

Layer	Checked statement	Status
Curvature	exact split, radius, recurrence, criterion	conditional reduction
Exponent transport	stable support, affine state, cancellation	proved here
Joint $(3, 5)$ transport	moment annihilation and criterion	conditional reduction
Prime jump	migration, sharp window, $(12, 5, 15)$	verified finite instance
Three transport	sharp window and $(60, 12)$	verified finite instance
First harmonic	cosine gap forces a certificate	conditional reduction
Adjacent phases	squared chord separation forces a certificate	conditional reduction
Pivot anti-reconstruction	pairwise phase energy and exact window-pair equivalence	conditional reduction
Tropical signature	fixed precision has a completion	unconditional progress
LCM factor ideals	factor/anchor and finite shift algebra have a synthetic survivor	unconditional progress
Irrationality of S	follows from a named unbounded supply	open

The development proves no cofinal supply of curvature, prime-jump, joint, or three-transport certificates, and no cofinal first-harmonic gap. It therefore does not prove Erdős #249.

References

[1] P. Erdős, *On arithmetical properties of Lambert series*, J. Indian Math. Soc. (N.S.) **12** (1948), 63–66, archive scan.

[2] T. M. Apostol, *Introduction to Analytic Number Theory*, Undergraduate Texts in Mathematics, Springer, 1976, doi:10.1007/978-1-4757-5579-4.

[3] P. B. Borwein, *On the irrationality of certain series*, Math. Proc. Cambridge Philos. Soc. **112** (1992), no. 1, 141–146, doi:10.1017/S030500410007081X.

[4] P. Erdős and R. L. Graham, *Old and New Problems and Results in Combinatorial Number Theory*, Monographies de l'Enseignement Mathématique 28 (1980), 61–62, scan.

[5] P. Erdős, *On the irrationality of certain series: problems and results*, in *New Advances in Transcendence Theory*, ed. A. Baker, Cambridge University Press, 1988, pp. 102–109, doi:10.1017/CBO9780511897184.009.

[6] T. Bloom (ed.), *Erdős Problems*, entry 249, `erdosproblems.com/249`, accessed July 2026.

[7] W. Cook, *Tail Certificates and Achievement-Set Geometry for Erdős Problems 249 and 257*, gateway exposition in this release.

[8] W. Cook, *Claim-Faithful Publication of a Formalised Mathematics Corpus*, systems companion in this release.